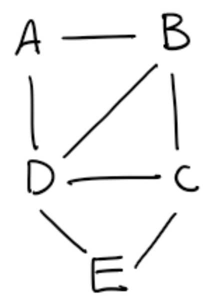
L5: Knowledge Check

1 Multiple answer 1 point

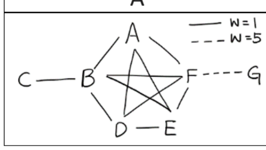
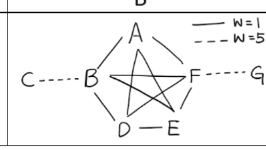
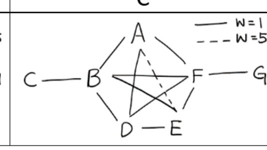
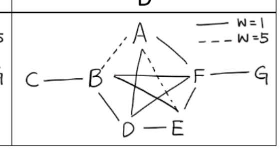
Which of the following clustering coefficient values is/are Correct?



- [ ] C(A) = 2/3
- [ ] C(B) = 2/3
- [ ] C(D) = 1/2
- [ ] C(entire network) = 11/15

2 Multiple choice 1 point

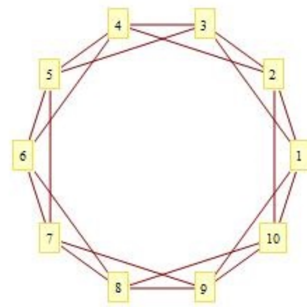
In which of the following networks, the weighted clustering coefficient value of node B is larger compared to the unweighted clustering coefficient value of the same node?

A	B	C	D
			

- ( ) A
- ( ) B
- ( ) C
- ( ) D

3 Multiple answer 1 point

Which of the following statement is/are True?



- [ ] The diameter of the above graph is 4
- [ ] The diameter of a ring network on n vertices increases linearly with the number of nodes
- [ ] The diameter of a clique increases linearly with the number of nodes
- [ ] The diameter of a complete binary tree increases logarithmically with the number of nodes

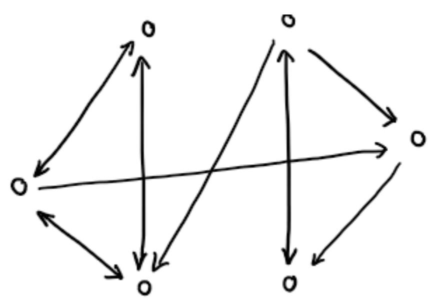
4 Multiple answer 1 point

Which of the following statement about the Watts-Strogatz (WS) and Preferential Attachment (PA) models is/are False?

- [ ] The WS model can have a power-law degree distribution by choosing an appropriate value for the rewiring probability
- [ ] The growth of CPL in power-law networks model, when the exponent gamma < 2, is very slow or zero
- [ ] The clustering coefficient of the WS model depends on the rewiring probability
- [ ] A PA network cannot have strong clustering with short paths but can have strong clustering with long paths

5 Multiple choice 1 point

How many “mutual out” motifs are in the below network?



- ○ 0
- ○ 1
- ○ 2
- ○ 3

6 Multiple answer 1 point

The following statements refer to network motifs in different types of networks. According to Lesson-5, which of the following statement(s) is/are TRUE?

- [ ] A common 3-node motif in gene regulatory networks is the FeedForward Loop (FFL)
- [ ] A common 3-node motif in electronic circuits (excluding forward logic chips) is the FeedBack Loop (FBL)
- [ ] A common 3-node motif in the WWW is the Clique
- [ ] A common 3-node motif in the neuronal networks (such as C.Elegans) is the FeedForward Loop (FFL)